curriculum vitae

Mauricio Sousa

マウリシオ・ソウサ

Associate Professor

Keio University Graduate School of Media Design

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google scholar

Research Interests

Human-Computer Interaction, Human-Robot Interaction, and Human-AI Interaction · Remote Collaboration, Telepresence, and Telexistence · Intelligent Environments and Spatial Computing · Mixed and Cross-Reality Environments · AI-Driven Interfaces, Wearables, and Embodied Companions · Haptic Communication and Multimodal Sensing · Robotic Avatars and Cyberphysical Augmentation

Short bio

I am an Associate Professor at the Keio University Graduate School of Media Design, affiliated with the Embodied Media Project. My research sits at the intersection of Human-Computer Interaction, spatial computing, and AI-driven interfaces. I design new interaction techniques that extend human perception, collaboration, and action across physical and virtual environments. Previously, I was a Senior Research Scientist at Reality Labs Research at Meta in Toronto, working on AI Wearables. Prior to that, I was a Postdoctoral Fellow at the Dynamic Graphics Project Lab (DGP), Department of Computer Science at the University of Toronto, working with Professor Tovi Grossman. Earlier, I was a researcher at the Visualization and Intelligent Multimodal Interfaces Group (VIMMI), University of Lisbon, under the supervision of Professor Joaquim Jorge. I hold a Ph.D., M.Sc., and B.Sc. in Computer Science and Engineering from Instituto Superior Técnico, University of Lisbon.

I explore how multimodal sensing, computer vision, and vision-language models can be embedded into wearable and spatial interfaces to support real-time interaction and communication. A core emphasis is enabling telepresence, remote collaboration, and context-aware guidance through AI-augmented AR/VR systems. From perception manipulation in 3D remote workspaces to haptic grammars for embodied AI companions, my work spans domains including engineering, healthcare, education, and everyday interaction. I have published over 50 peer-reviewed papers in top-tier venues such as ACM CHI, UIST, ISS, IUI, VRST, IEEE VR, IJHCS, and IEEE TVCG, and received multiple awards recognizing research excellence.

Education

Ph.D. in Computer Science and Engineering **2020**
Instituto Superior Técnico, University of Lisbon, Portugal
Thesis: *Perception Manipulation for Seamless Face-to-face Remote Collaboration*
Summa Cum Laude (highest honor for Doctoral Degree)
Advisor: Prof. Joaquim Jorge
Committee: Prof. Anthony Steed, Prof. Pedro Campos, Prof. Carlos Martinho, Prof. Miguel Sales Dias, and Prof. Pavão Martins

M.Sc. in Information Systems and Computer Engineering **2014**
Instituto Superior Técnico, University of Lisbon, Portugal
Thesis: *Remote Proxemics for Collaborative Virtual Environments*
Advisors: Prof. Joaquim Jorge and Prof. Alfredo Ferreira
Committee: Prof. Carlos Duarte and Prof. Pedro Sousa

B.Sc. in Information Systems and Computer Engineering **2012**
Instituto Superior Técnico, University of Lisbon, Portugal

Experience

Graduate School of Media Design, Keio University (KMD), Japan
慶應義塾大学大学院メディアデザイン研究科 **Apr 2026 - Present**

Associate Professor – Lecturing on Interaction Design and Sketching User Experiences. Also, conducting research and mentoring at the intersection of intelligent spaces, spatial computing, and Human-Computer Interaction.

- Leading research on intelligent environments, embodied media, and AI-powered wearable devices for multimodal interaction across physical and virtual spaces in collaboration with the Embodied Media Project and Cybernetic Being Project.
- Teaching graduate lectures in Human-Computer Interaction, Interaction Design, and Sketching User Experiences.
- Mentoring graduate students and collaborating on research that integrates spatial computing, Human-Robot Interaction, tangible interfaces, and AI-driven accessibility across mixed reality environments.

Graduate School of Media Design, Keio University (KMD), Japan
慶應義塾大学大学院メディアデザイン研究科 **Oct 2025 - Mar 2026**

Project Senior Assistant Professor – Conducting research, teaching, and mentoring in Human-Computer Interaction, AI wearables, spatial computing, and mixed reality interaction.

- Leading research on embodied media, AI-powered wearable devices, and multimodal interaction in collaboration with the Embodied Media Project and Cybernetic Being Project.
- Teaching and mentoring graduate students within KMD's interdisciplinary design and innovation framework.

- Collaborating on research that integrates Human-Computer Interaction, Human-Robot Interaction, tangible interfaces, AI communication, and accessibility across physical and virtual spaces.

Meta Reality Labs Research, Toronto — 2024

Senior Research Scientist – Researching, designing, and evaluating advanced input interaction techniques for wearables, AI, and the next generation of neural interfaces and spatial computing.

- Led the planning, prototyping, and user evaluation of learning and onboarding approaches for neural interfaces and spatial computing gesture input.
- Designed and evaluated novel AI wearable devices for input and haptics, with a focus on AI companion communication and multimodal interaction.

DGP Lab, Dept. of Computer Science, University of Toronto — 2020 - 2024

Postdoctoral Fellow – Researching, designing, and evaluating novel interaction techniques. Mentoring and supporting undergraduate and graduate students.

- Led and collaborated on projects involving novel interaction techniques, AI-driven creativity tools, wearable I/O, and spatial computing.
- Mentored 13 undergraduate and graduate students and co-advised 2 Master's theses on topics such as animated storytelling in MR and physiological sensing in education.
- Contributed to research on telepresence, multimodal haptics, generative interaction design, and embodied AI sensing for remote and immersive collaboration.
- Co-authored over 15 peer-reviewed publications, including in *ACM CHI*, *UIST*, *IUI*, *IEEE VR*, and *ISMAR*.
- Received multiple awards, including CHI and UIST Honorable Mentions, and the Jazzki 2024 Human-Machine Creativity Award.

Champalimaud Center for the Unknown — 2018 - 2020

Researcher – Researching, designing, and evaluating novel mixed reality interaction techniques for minimally invasive laparoscopic surgery.

- Developed augmented reality systems to assist with anatomical understanding and surgical planning.
- Developed Laparoscopic simulation hardware for training with augmented reality heads-up display capabilities.
- Collaborated with surgeons and medical experts to ensure clinical relevance and usability.
- Co-authored 2 publications on laparoscopic surgery and anatomical visualization, including articles in *Journal of Biomedical Informatics* and *Computers & Graphics*.

INESC-ID, Visualization and Intelligent Multimodal Interfaces — 2012 - 2020

Researcher – Researching, designing, and evaluating novel interaction techniques. Managing and maintaining VIMMI's media lab.

- Researched 3D user interfaces, spatial computing, and AR/VR techniques for collocated and remote collaboration.

- Developed interactive systems using head-mounted displays, tabletops, wearables, and large-scale displays.
- Co-authored more than 25 peer-reviewed publications in CHI, IEEE VR, VRST, and TVCG.
- Co-advised students on projects in mixed reality, proxemics, virtual anatomy, and haptics.
- Contributed to funded research projects on architectural modeling, medical visualization, and collaborative industrial design.

Publications

Throughout my academic and industry career, I have built a multidisciplinary research portfolio at the intersection of Human-Computer Interaction, spatial computing, and embodied interaction. My work appears regularly in the leading venues of the field, including ACM CHI, ACM UIST, IEEE ISMAR, and IEEE VR,[1] and has been recognized with a Best Paper Award at ACM VRST, Honorable Mention Awards at ACM CHI, and the Jazzki Human-Machine Creativity Award, as well as media coverage.

Overview:
36 conference proceedings, **8** journal articles, **2** book chapters, **19** other peer-reviewed publications, and **2** theses.
1880+ citations, h-index **24** (Google Scholar).

Peer-reviewed Conference Proceedings

C36 **ITO: Real-time Browser Tab Orchestration Through Intent Detection.** Elvin Hu, Jasmine Sumpter, Takatoshi Yoshida, Mauricio Sousa, and Kouta Minamizawa. ACM Symposium on User Interface Software and Technology (UIST), 2026

C35 **Body Sublimation: Designing State-Changing Extended Hand for Out-of-Reach Object Manipulation.** Masahiro Ohta, Reigo Ban, Hideaki Nii, Guillaume Gourmelen, Takuji Narumi, Mauricio Sousa, Takatoshi Yoshida, and Kouta Minamizawa. ACM Symposium on User Interface Software and Technology (UIST), 2026

C34 **CapAura: A Proxemic Interactive Infrastructure on Everyday Furniture through Capacitive Architecture Unit.** Jianan Zhang, Kohei Kobayashi, Hanrui Peng, Hideaki Nii, Takatoshi Yoshida, Mauricio Sousa, and Kouta Minamizawa. ACM Symposium on User Interface Software and Technology (UIST), 2026

C33 **Adaptique: Multi-objective and Context-aware Online Adaptation of Selection Techniques in Virtual Reality.** Chao-Jung Lai, Mauricio Sousa, Tianyu Zhang, Ludwig Sidenmark, and Tovi Grossman ACM Symposium on User Interface Software and Technology (UIST), 2025

C32 **VibraForge: A Scalable Prototyping Toolkit For Creating Spatialized Vibrotactile Feedback Systems.** Bingjian Huang, Siyi Ren, Yuewen Luo, Qilong Cheng, Hanfeng Cai, Yeqi Sang, Mauricio Sousa, Paul H. Dietz, and Daniel Wigdor. ACM Conference on Human Factors in Computing Systems (CHI), 2025

C31 **Sparc: Shared perspective with avatar distortion for remote collaboration in VR.** João Simões, Anderson Maciel, Catarina Moreira, Mauricio Sousa, and Joaquim Jorge Computer

[1] **Note on publication venues:** in Human-Computer Interaction, ACM CHI, ACM UIST, IEEE ISMAR, and IEEE VR are widely regarded as the premier forums for disseminating research results, and together they cover the broad spectrum of the field.

Graphics International Conference (CGI), 2024

C30 **Exploring User Placement for VR Remote Collaboration in a Constrained Passenger Space.** Daniel Medeiros, Graham Wilson, Mauricio Sousa, Nadia Pantidi, Diego Drago, Mark McGill, and Stephen Brewster. ACM Symposium on Virtual Reality Software and Technology (VRST), 2024

Best Paper Award

C29 **PhoneInVR: An Evaluation of Spatial Anchoring and Interaction Techniques for Smartphone Usage in Virtual Reality.** Fengyuan Zhu, Mauricio Sousa, Ludwig Sidenmark, and Tovi Grossman. ACM Conference on Human Factors in Computing Systems (CHI), 2024.

C28 **SynthScribe: Deep Multimodal Tools for Synthesizer Sound Retrieval and Exploration.** Stephen Brade, Bryan Wang, Mauricio Sousa, Gregory Lee Newsome, Sageev Oore, and Tovi Grossman. ACM Conference on Intelligent User Interfaces (IUI), 2024

Jazzki 2024 Human-Machine Creativity Award

C27 **PinchLens: Applying Spatial Magnification and Adaptive Control Display Gain for Precise Selection in Virtual Reality.** Fengyuan Zhu, Ludwig Sidenmark, Mauricio Sousa, and Tovi Grossman. IEEE International Symposium on Mixed and Augmented Reality (ISMAR), 2023

C26 **Promptify: Text-to-Image Generation through Interactive Prompt Exploration with Large Language Models** Stephen Brade, Bryan Wang, Mauricio Sousa, Sageev Oore, and Tovi Grossman. ACM Symposium on User Interface Software and Technology (UIST), 2023

C25 **EnchantedBrush: Animating in Mixed Reality for Storytelling and Communication** Eve Mingxiao Li, Anran Qi, Mauricio Sousa, and Tovi Grossman. ACM GI Graphics Interface Conference, 2023.

C24 **Stargazer: An Interactive Camera Robot for Capturing How-To Videos Based on Subtle Instructor Cues** Jiannan Li, Mauricio Sousa, Karthik Mahadevan, Bryan Wang, Paula Akemi Aoyagui, Nicole Yu, Angela Yang, Ravin Balakrishnan, Anthony Tang, and Tovi Grossman. ACM Conference on Human Factors in Computing Systems (CHI), 2023.

C23 **Investigating Guardian Awareness Techniques to Promote Safety in Virtual Reality.** Sixuan Wu, Jiannan Li, Mauricio Sousa, and Tovi Grossman. IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR), 2023.

C22 **MAGIC: Manipulating Avatars and Gestures to Improve Remote Collaboration.** Catarina G. Fidalgo, Mauricio Sousa, Daniel Mendes, Rafael dos Anjos, Daniel Medeiros, Karan Singh, and Joaquim Jorge. IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR), 2023.

C21 **Touching The Droid: Understanding and Improving Touch Precision With Mobile Devices in Virtual Reality.** Fengyuan Zhu, Zhuoyue Lyu, Mauricio Sousa, and Tovi Grossman. IEEE International Symposium on Mixed and Augmented Reality (ISMAR), 2022

C20 **immersivePOV: Filming How-To Videos with a Head-Mounted 360° Action Camera.** Kevin Huang, Jiannan Li, Mauricio Sousa, and Tovi Grossman. ACM Conference on Human Factors in Computing Systems (CHI), 2022.

Honorable Mention Award

C19 **ASTEROIDS: Exploring Swarms of Mini-Telepresence Robots for Physical Skill Demonstration.** Jiannan Li, Mauricio Sousa, Chu Li, Jessie Liu, Yan Chen, Ravin Balakrishnan, and Tovi Grossman. ACM Conference on Human Factors in Computing Systems (CHI), 2022.

C18 **Route Tapestries: Navigating 360° Virtual Tour Videos Using Slit-Scan Visualizations.** Jiannan Li, Jiahe Lyu, Mauricio Sousa, Ravin Balakrishnan, Anthony Tang, and Tovi Grossman.

ACM Symposium on User Interface Software and Technology (UIST), 2021

C17 **Promoting Reality Awareness in Virtual Reality through Proxemics.** Daniel Medeiros, Rafael dos Anjos, Nadia Pantidi, Kun Huang, Mauricio Sousa, Craig Anslow and Joaquim Jorge. IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR), 2021.

C16 **"Grip-that-there": An Investigation of Explicit and Implicit Task Allocation Techniques for Human-Robot Collaboration.** Karthik Mahadevan, Mauricio Sousa, Anthony Tang, and Tovi Grossman. ACM Conference on Human Factors in Computing Systems (CHI), 2021.
Honorable Mention Award

C15 **Negative Space: Investigating Workspace Awareness in 3D Face-to-face Remote Collaboration.** Mauricio Sousa, Daniel Medeiros, and Joaquim Jorge. ACM SIGGRAPH International Conference on Virtual-Reality Continuum and its Applications in Industry (VRCAI), 2019

C14 **Safe Walking in VR.** Mauricio Sousa, Daniel Medeiros, Francisco Venda, and Joaquim Jorge. ACM SIGGRAPH International Conference on Virtual-Reality Continuum and its Applications in Industry (VRCAI), 2019

C13 **Adventures in Hologram Space: Exploring the Design Space of Eye-to-eye Volumetric Projection-based Telepresence.** Rafael Kuffner dos Anjos, Mauricio Sousa, Daniel Medeiros, Daniel Mendes, Mark Billinghurst, Craig Anslow and Joaquim Jorge. ACM Symposium on Virtual Reality Software and Technology (VRST), 2019

C12 **WARPING DEIXIS: Distorting Gestures to Enhance Collaboration.** Mauricio Sousa, Rafael Kuffner Dos Anjos, Daniel Mendes, Mark Billinghurst, and Joaquim Jorge. ACM Conference on Human Factors in Computing Systems (CHI), 2019
Featured in the 'Best of CHI 2019' event by IndiaHCI

C11 **Using Custom Transformation Axes for Mid-Air Manipulation of 3D Virtual Objects.** Daniel Mendes, Mauricio Sousa, Rodrigo Lorena, Alfredo Ferreira, and Joaquim Jorge. ACM Symposium on Virtual Reality Software and Technology (VRST), 2017

C10 **Creepy Tracker Toolkit for Context-aware Interfaces.** Mauricio Sousa, Daniel Mendes, Rafael Kuffner dos Anjos, Daniel Medeiros, Alberto Raposo, Alfredo Ferreira, João Pereira, and Joaquim Jorge. ACM Interactive Surfaces and Spaces (ISS), 2017

C9 **VRRRRoom: Virtual Reality for Radiologists in the Reading Room.** Mauricio Sousa, Daniel Mendes, Soraia Paulo, Nuno Matela, Joaquim Jorge, and Daniel S. Lopes. ACM Conference on Human Factors in Computing Systems (CHI), 2017

C8 **PRECIOUS! Out-of-reach Selection using Iterative Refinement in VR.** Daniel Mendes, Daniel Medeiros, Eduardo Cordeiro, Mauricio Sousa, Alfredo Ferreira, and Joaquim Jorge. IEEE Symposium on 3D User Interfaces (3DUI), 2017

C7 **Mid-air Modeling with Boolean Operations in VR.** Daniel Mendes, Daniel Medeiros, Mauricio Sousa, Ricardo Ferreira, Alberto Raposo, Alfredo Ferreira, and Joaquim Jorge. IEEE Symposium on 3D User Interfaces (3DUI), 2017

C6 **Effects of Speed and Transitions on Target-based Travel Techniques.** Daniel Medeiros, Eduardo Cordeiro, Daniel Mendes, Mauricio Sousa, Alberto Raposo, Alfredo Ferreira and Joaquim Jorge. ACM Symposium on Virtual Reality Software and Technology (VRST), 2016

C5 **Perceiving Depth: Optical versus Video See-through.** Daniel Medeiros, Mauricio Sousa, Daniel Mendes, Alberto Raposo, and Joaquim Jorge. ACM Symposium on Virtual Reality Software and Technology (VRST), 2016

C4 **SleeveAR: Augmented Reality for Rehabilitation using Realtime Feedback.** Mauricio Sousa, João Vieira, Daniel Medeiros, Artur Arsenio, and Joaquim Jorge. ACM Intelligent User

Interfaces (IUI), 2016

C3 **From Tecton to Teknos: Going back to the Classical Roots of Architecture.** Daniel Mateus, Mauricio Sousa, Rui de Klerk, Sandra Gama, Joaquim Jorge, and José Duarte. Education and Research in Computer Aided Architectural Design in Europe (eCAADe), 2015

C2 **Eery Space: Facilitating Virtual Meetings Through Remote Proxemics.** Mauricio Sousa, Daniel Mendes, Alfredo Ferreira, João Madeiras Pereira, and Joaquim Jorge. Human-Computer Interaction—INTERACT, 2015

C1 **Beyond Post-It: Structured Multimedia Annotations for Collaborative VEs.** João Guerreiro, Daniel Pires, Mauricio Sousa, Daniel Mendes, Ismael Santos, Alberto Raposo, and Joaquim Jorge. Eurographics Symposium on Virtual Environments (EGVE), 2014

Journal Articles

J8 **Exploring AR Hand Augmentations as Error Feedback Mechanisms for Enhancing Gesture-based Tutorials** Catarina G. Fidalgo, Yukang Yan, Mauricio Sousa, Joaquim Jorge, and David Lindlbauer. Frontiers in Virtual Reality, 2025

J7 **A Survey on Remote Assistance and Training in AR and VR Environments**. Catarina G. Fidalgo, Yukang Yan, Hyunsung Cho, Mauricio Sousa, David Lindlbauer, and Joaquim Jorge. IEEE Transactions on Visualization and Computer Graphics (TVCG), 2023

J6 **Laparoscopy with Augmented Reality Adaptations.** Ezequiel Zorzal, José Miguel Gomes, Mauricio Sousa, Pedro Belchior, Pedro G da Silva, Nuno Figueiredo, Daniel S. Lopes, and Joaquim Jorge. Elsevier Journal of Biomedical Informatics, 2020

J5 **Anatomy Studio: a Tool for Virtual Dissection Through Augmented 3D Reconstruction Sessions.** Ezequiel Zorzal, Mauricio Sousa, Daniel Mendes, Rafael K dos Anjos, Soraia F. Paulo, Pedro Rodrigues, José Mendes, Vincent Delmas, Jean-Francois Uhl, José Mogorrón, Daniel S. Lopes, and Joaquim Jorge. Computers & Graphics, 2019

J4 **Magic Carpet: Interaction Fidelity for Flying in VR.** Daniel Medeiros, Mauricio Sousa, Alberto Raposo, and Joaquim Jorge. IEEE Transactions on Visualization and Computer Graphics (TVCG), 2019

Recipient of the Encarnação Award 2020 from Eurographics Portuguese chapter

J3 **Design and evaluation of novel out-of-reach selection techniques for VR using iterative refinement.** Daniel Mendes, Daniel Medeiros, Mauricio Sousa, Eduardo Cordeiro, Alfredo Ferreira, and Joaquim Jorge. Computers & Graphics, 2017

Honourable Mention in the 33rd Spring Conference on Computer Graphics

J2 **Hip-directed walking-in-place using a single depth camera.** Luís Bruno, Mauricio Sousa, Alfredo Ferreira, João Madeiras Pereira, and Joaquim Jorge. International Journal of Human-Computer Studies (IJHCS), Elsevier, 2017

J1 **Expeditious Illustration of Layer-Cake Models On and Above a Tactile Surface.** Daniel S. Lopes, Daniel Mendes, Mauricio Sousa, and Joaquim Jorge. Computers & Geosciences (in press), 2016

Book Chapters

B2 **A Tool for Collaborative Anatomical Dissection.** Ezequiel Zorzal, Mauricio Sousa, Daniel Mendes, Soraia Paulo, Pedro Rodrigues, Joaquim Jorge, and Daniel Simões Lopes. Book

chapter in Digital Anatomy, Springer, 2021

B1 **Remote Proxemics.** Mauricio Sousa, Daniel Mendes, Daniel Medeiros, Alfredo Ferreira, João Madeiras Pereira, and Joaquim Jorge. Book chapter in Collaboration Meets Interactive Spaces, Springer, 2016

Other Peer-Reviewed Publications

019 **Exploring Gradual Narrative Transitions to Avoid Audience Exclusion in XR storytelling.** Jasmine Sumpter, Jada Sumpter, Takatoshi Yoshida, Kouta Minamizawa, and Mauricio Sousa. The 32nd ACM Symposium on Virtual Reality Software and Technology (VRST), 2026

018 **Noticing What We Miss: A Preliminary Study on Delegating the Act of Noticing to a VLM-Guided Companion Sidekick.** Cassie Zhu, Takatoshi Yoshida, Mauricio Sousa, and Kouta Minamizawa. The 31st Annual Meeting of the Virtual Reality Society of Japan (VRSJ), 2026

017 **A Preliminary Exploration of Gradual Transitions into Beyond-Human XR Narratives.** Jasmine Sumpter, Takatoshi Yoshida, Mauricio Sousa, and Kouta Minamizawa. The 31st Annual Meeting of the Virtual Reality Society of Japan (VRSJ), 2026

016 **A Preliminary Study on an Audio-Haptic Augmentation System Using Body Orientation as a Cue for Supporting Bidirectional Conversation in Extended Spaces.** Xuan Chen, Shen Jiayi, Mauricio Sousa, and Kouta Minamizawa. The 31st Annual Meeting of the Virtual Reality Society of Japan (VRSJ), 2026

015 **Body Sublimation: XRとロボットによる身体化感覚を保った物体操作のための身体レンダリング手法.** Masahiro Ohta, Reigo Ban, Hideaki Nii, Guillaume Gourmelen, Takuji Narumi, Mauricio Sousa, Takatoshi Yoshida, and Kouta Minamizawa. The 31st Annual Meeting of the Virtual Reality Society of Japan (VRSJ), 2026

014 **Experience Sharing Studio: An Immersive Platform for Spatio-Temporal Recording and Playback of Embodied Multisensory Memory.** H. Higashi, Y. Zhu, K. Kobayashi, K. Tsujita, J. Morita, E. Koizumi, J. Shen, Mauricio Sousa, A. Horie, Takatoshi Yoshida, and Kouta Minamizawa. ACM Special Interest Group on Computer Graphics and Interactive Techniques Conference Emerging Technologies (SIGGRAPH), 2026

013 **VRChoir: Exploring Remote Choir Rehearsals via Virtual Reality** Tianquan Di, Daniel Medeiros, Mauricio Sousa, and Tovi Grossman. IEEE Conference on Virtual Reality and 3D User Interfaces Posters (IEEE VR), 2023

012 **Anatomy Studio II: A Cross-Reality Application for Teaching Anatomy.** Joaquim Jorge, Pedro Belchior, Abel Gomes, Mauricio Sousa, João Pereira, and Jean-François Uhl. XR for Healthcare and Wellbeing Workshop (IEEE VR), 2022

011 **Design requirements to improve laparoscopy via XR.** Ezequiel R Zorzal, Mauricio Sousa, Pedro Belchior, João Madeiras Pereira, Nuno Figueiredo, and Joaquim Jorge. XR for Healthcare and Wellbeing Workshop (IEEE VR), 2022

010 **Constellation: a Multi-User Interface for Remote Drone Tours.** Jiannan Li, Mauricio Sousa, Ravin Balakrishnan, and Tovi Grossman. International Conference on Human-Agent Interaction (HAI), 2021

09 **Demo hour.** Paden Shorey, Audrey Girouard, Sang Ho Yoon, Yunbo Zhang, Ke Huo, Karthik Ramani, Mauricio Sousa, Daniel Mendes, Soraia Paulo, Nuno Matela, Joaquim Jorge, Daniel S. Lopes, Dirk Wenig, Johannes Schöning, Alex Olwal, Mathias Oben, and Rainer Malaka. Demo hour. interactions 24, 6 (October 2017), 8-11.

08 **Evaluation of Travel Techniques for Virtual Reality.** Eduardo Cordeiro, Daniel Medeiros, Daniel Mendes, Mauricio Sousa, Alberto Raposo, Alfredo Ferreira, and Joaquim Jorge. Portuguese Meeting of Computer Graphics (EPCG), 2016

07 **Beyond Eery Space: Applying Gradual Engagement to Remote Proxemics.** Mauricio Sousa, Daniel Medeiros, Alberto Raposo, and Joaquim Jorge. Collaboration meets Interactive Surfaces Workshop, ACM Interactive Tabletops and Surfaces(ITS), 2015

06 **Augmented Reality for Rehabilitation Using Multimodal Feedback.** João Vieira, Mauricio Sousa, Artur Arsénio, and Joaquim Jorge. REHAB2015 Workshop, 2015

05 **Enabling Remote Proxemics through Multiple Surfaces.** Daniel Mendes, Mauricio Sousa, João Madeiras Pereira, Alfredo Ferreira, and Joaquim Jorge. Collaboration meets Interactive Surfaces Workshop, ACM Interactive Tabletops and Surfaces(ITS), 2014

04 **Eery Proxemics: Proximidade à Distância usando Múltiplas Superfícies.** Mauricio Sousa, Daniel Mendes, João Madeiras Pereira, Alfredo Ferreira, and Joaquim Jorge. Portuguese Meeting of Computer Graphics (EPCG), 2014

03 **ThumbCam: Returning to single touch interactions to explore 3D virtual environments.** Daniel Mendes, Mauricio Sousa, Alfredo Ferreira, and Joaquim Jorge. ACM Interactive Tabletops and Surfaces(ITS), 2014

02 **Binding a Handheld Device with its Owner. Mauricio Sousa and Joaquim Jorge.** Collaboration meets Interactive Surfaces Workshop, ACM Interactive Tabletops and Surfaces (ITS), 2013

01 **Collaborative 3D Visualization on Large Screen Displays.** Daniel Mendes, Mauricio Sousa, Bruno Araújo, Alfredo Ferreira, Hildegardo Noronha, Pedro Campos, Luciano Soares, Alberto Raposo, and Joaquim Jorge. Powerwall Workshop, SIGCHI Conference on Human Factors in Computing Systems (CHI), 2013

Theses

T2 **Perception Manipulation for Seamless Face-to-face Remote Collaboration.**
Mauricio Sousa. Ph.D. Thesis. Instituto Superior Técnico, University of Lisbon. 2020

T1 **Remote Proxemics for Collaborative Virtual Environments.**
Mauricio Sousa. M.Sc. Thesis. Instituto Superior Técnico, University of Lisbon. 2014

Academic Service

Program Committee

ACM Conference on Human Factors in Computing Systems (CHI)
Associate Chair
2027
ACM Conference on Human Factors in Computing Systems (CHI)
Subcommittee Associate Chair: *Interacting with Devices: Interaction Techniques & Modalities*
2024, 2025, 2026
ACM Symposium on User Interface Software and Technology (UIST)

Associate Chair
2024, 2026
ACM Conference on Human Factors in Computing Systems (CHI)
Awards Committee Member for the subcommittee: *Interacting with Devices: Interaction Techniques & Modalities*
2025
IEEE International Symposium on Mixed and Augmented Reality (ISMAR)
2021, 2024, 2025, 2026
ACM Conference on Designing Interactive Systems (DIS)
Interactivity (Demonstrations) Jury
2026
ACM International Conference on Multimodal Interaction (ICMI)
2020

Conference Organizing Committee

5 ACM Conference on Tangible, Embedded, and Embodied Interaction (TEI)
Student Design Competition Co-chair
2027

4 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)
Video Co-chair
2023

3 ACM International Conference on Interactive Surfaces and Spaces (ISS)
Web Chair and Online Experience Co-chair
2020, 2022

2 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)
Web Chair
2021, 2022

1 Eurographics
Fast-forward Chair
2016

Workshops Organizing Committee

W2 **Reimagining Human Augmentation through Disability-Led Performance.** Giulia Barbareschi, Ximing Shen, Ryoichi Ando, Mark Armstrong, Mauricio Sousa, and Kouta Minamizawa. Augmented Humans International Conference (AHs), 2026

W1 **Enhancing cross-reality applications and user experiences.** Frank Maurer, Craig Anslow, Joaquim Jorge, and Mauricio Sousa. International Conference on Advanced Visual Interfaces (AVI), 2022

Conference Session Chair

3 ACM Symposium on User Interface Software and Technology (UIST)
Session: *Shaping and Sharing Spaces in XR*

2026

2 ACM Conference on Human Factors in Computing Systems (CHI)
Session: *Haptic and Multisensory Feedback*
2026

1 ACM Symposium on User Interface Software and Technology (UIST)
Session: *Device Augmentation & Communication*
2021

Peer Reviewer

24 ACM Conference on Human Factors in Computing Systems (CHI)
2018, 2020, 2021, 2022, 2023, 2024

23 ACM Symposium on User Interface Software and Technology (UIST)
2020, 2021, 2022, 2023

22 ACM Symposium on Virtual Reality Software and Technology (VRST)
2017, 2018, 2020, 2023, 2024

21 Transactions on Visualization and Computer Graphics (TVCG)
2023, 2024, 2025, 2026

20 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)
2019, 2023, 2024, 2026

19 Augmented Humans (AHs) International Conference (AHs2026)
2026

18 International Conference on Graphics and Interaction (ICGI)
2021, 2022, 2023, 2024, 2025, 2026

17 ACM International Conference on Multimodal Interaction (ICMI)
2018, 2019, 2024

16 Human-Computer Interaction (HCI)
2024

15 International Journal of Human-Computer Interaction (IJHCI)
2025

14 The International Journal of Human-Computer Studies (IJHCS)
2023

13 ACM International Conference on Tangible, Embedded, and Embodied Interaction (TEI)
2018, 2020, 2023, 2026

12 IEEE VIS
2023

11 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)
2017, 2018, 2019, 2020, 2021, 2022, 2023

10 ACM Conference on Designing Interactive Systems (DIS)
2016, 2021, 2023

9 MDPI Sensors
2023

8 Springer Virtual Reality (VR)
2023

7 ACM Conference on Intelligent User Interfaces (IUI)

2023

6 Elsevier Computers & Graphics
2018, 2019, 2020, 2021, 2023

5 International Conference on Graphics and Interaction (ICGI)
2021, 2022, 2023, 2025

4 ACM International Conference on Interactive Surfaces and Spaces (ISS)
2015, 2016, 2017, 2018, 2019, 2020, 2022, 2025

3 ACM Symposium on Spatial User Interaction (SUI)
2017, 2020

2 ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW)
2020

1 IEEE Consumer Electronics Magazine (CEM)
2016

Thesis & Dissertation Committees

9 Doctoral Dissertation Proposal Committee for **Kinga Skiers** **2026**
Keio University Graduate School of Media Design (KMD)
Co-designing Mental Health Technologies with Neurodivergent People and Clinicians
Role: Co-advisor (Principal Advisor: Prof. Kouta Minamizawa; Co-advisor: Prof. Dunya Donna Chen)

8 Doctoral Dissertation Proposal Committee for **Danyang Peng** **2026**
Keio University Graduate School of Media Design (KMD)
Designing Body-Centered Meditative Interactions through Sequential Vibrotactile Guidance
Role: Co-advisor (Principal Advisor: Prof. Kouta Minamizawa; Co-advisor: Prof. Chihiro Sato)

7 Doctoral Dissertation Proposal Committee for **Carlos Garcia Fernandes** **2026**
Keio University Graduate School of Media Design (KMD)
Designing Ephemeral Layers: A Systematic Approach to Composing Spatial Effects Through Technological Mediation in Architecture
Role: Co-advisor (Principal Advisor: Prof. Kouta Minamizawa; Co-advisors: Prof. Matthew Waldman, and Prof. Yamaoka)

6 Master's Thesis Committee (Reviewer) for **He Xianzhi** **2026**
Keio University Graduate School of Media Design (KMD)
Thesis: *Designing for Creative Re-engagement: An Interactive Tool for Overcoming Creative Blocks in Digital Illustration*

5 Master's Thesis Committee (Reviewer) for **Komoriya Yumi** **2026**
Keio University Graduate School of Media Design (KMD)
Thesis: *The Inspiration Sheet and the Catch Cards: Designing a Language Toolkit to Cultivate Creative Confidence*

4 Master's Thesis Committee (Reviewer) for **Ke Xu** **2026**
Keio University Graduate School of Media Design (KMD)
Thesis: *From Rules to Reasons: Interactive Etiquette Learning for Chinese Tourists in Japan*

3 Master's Thesis Committee (Reviewer) for **Shang Mei Cheng** **2026**
Keio University Graduate School of Media Design (KMD)
Thesis: *Enhancing AR Digital Narrative Tourism Experiences from the Perspective of the Experience*

Economy: A Case Study of Otome Games

2 Master's Thesis Committee (Reviewer) for **Lo Bok Yin** **2026**
Keio University Graduate School of Media Design (KMD)
Thesis: *Designing Participant-Driven Spatial Narratives: "Enhancing Engagement and Curiosity in Cultural Exhibitions"*

1 Master's Thesis Committee for **Filipe Guedes Barbosa** **2022**
Faculdade de Engenharia, Universidade do Porto
Thesis: *Shape-a-getti: A haptic device for getting multiple shapes using a single actuator.*

Supervision, and Mentoring

PhD Students

5 **Hua Ma**, Keio University Graduate School of Media Design (KMD) **2026 -**
Co-Advisor with Junichi Yamaoka.
Thesis: *TBD*

4 **Adam Kadmani**, Keio University Graduate School of Media Design (KMD) **2026 -**
Co-Advisor with Kouta Minamizawa.
Thesis: *TBD*

3 **Kinga Skiers**, Keio University Graduate School of Media Design (KMD) **2026 -**
Co-Advisor with Kouta Minamizawa.
Thesis: *TBD*

2 **Peng Danyang (Phoebe)**, Keio University Graduate School of Media Design (KMD) **2026 -**
Co-Advisor with Kouta Minamizawa.
Thesis: *TBD*

1 **Ismael Rasa**, Keio University Graduate School of Media Design (KMD) **2026 -**
Co-Advisor with Kouta Minamizawa.
Thesis: *TBD*

Master Students

21 **Rachel Ciara Nelson**, Keio University Graduate School of Media Design (KMD) **2026 -**
Co-Advisor with Junichi Yamaoka.
Thesis: *TBD*

20 **Yihan Guan**, Keio University Graduate School of Media Design (KMD) **2026 -**
Co-Advisor with Junichi Yamaoka.
Thesis: *TBD*

19 **Zechun Lin (Livia)**, Keio University Graduate School of Media Design (KMD) **2026 -**
Co-Advisor with Nanako Ishido.
Thesis: *TBD*

18 **Yu-Cih Lin (Pamelia)**, Keio University Graduate School of Media Design (KMD) **2026 -**

Thesis: *TBD*

17 **Xinyu Zhou (Zoey)**, Keio University Graduate School of Media Design (KMD) **2026 -**
Thesis: *TBD*

16 **Fernando Chavez Rivera**, Keio University Graduate School of Media Design (KMD) **2026 -**
Thesis: *TBD*

15 **Xuan Chen**, Keio University Graduate School of Media Design (KMD) **2025 -**
Co-Advisor with Kouta Minamizawa.
Thesis: *TBD*
[O16]

14 **Schweizer Rene Pascal**, Keio University Graduate School of Media Design (KMD) **2025 -**
Co-Advisor with Kouta Minamizawa.
Thesis: *TBD*

13 **Jasmine Sumpter**, Keio University Graduate School of Media Design (KMD) **2025 -**
Co-Advisor with Kouta Minamizawa.
Thesis: *TBD*
[C36] [O17] [O19]

12 **Cassie Zhu**, Keio University Graduate School of Media Design (KMD) **2025 -**
Co-Advisor with Kouta Minamizawa.
Thesis: *TBD*
[O18]

11 **Elvin Hu**, Keio University Graduate School of Media Design (KMD) **2025 -**
Co-Advisor with Kouta Minamizawa
Thesis: *TBD*
[C36]

10 **Yuze Tian (Uzy)**, Keio University Graduate School of Media Design (KMD) **2025 - 2026**
Co-Advisor with Junichi Yamaoka.
Thesis: *Sensor-Driven Kinetic Structures: Affective feedback through physical motion in everyday environments.*

9 **Xia Zicheng (Natsu)**, Keio University Graduate School of Media Design (KMD) **2025 - 2026**
Co-Advisor with Kouta Minamizawa and Takatoshi Yoshida.
Thesis: *DigiSnap: Rapid In Situ Digitization of Real-World Objects into Interactive Assets for Mixed Reality*

8 **Jiannan Zhang (Sonan)**, Keio University Graduate School of Media Design (KMD) **2025 - 2026**
Co-Advisor with Kouta Minamizawa and Takatoshi Yoshida.
Thesis: *Capacitive Skin: A Proxemic Interactive Infrastructure on Everyday Furniture through Capacitive Architecture Unit.*

7 **Mingxiao (Eve) Li**, University of Toronto. **2021 - 2023**
Co-Advisor with Tovi Grossman.
Thesis: *EnchantedBrush: Animating in Mixed Reality for Storytelling and Communication*
[C25]

6 **João Simões**, Instituto Superior Técnico, University of Lisbon. **2021 - 2023**

Co-Advisor with Joaquim Jorge.
Thesis: *SURI: Stretching User References for Interaction - Multi-user collaboration with a shared perspective*
[C31]

5 **Carlos McGregor Muro**, University of Toronto. **2020 - 2022**
Co-Advisor with Tovi Grossman.
Thesis: *Knowing when students struggle before verbalizing it using non-intrusive psycho-physiological indicators.*

4 **Manuel Lopes**, Instituto Superior Técnico, University of Lisbon. **2020 - 2021**
Co-Advisor with Joaquim Jorge.
Thesis: *CHASM - Computer-Human Assisted Segmentation of Medical Structures*

3 **Catarina G. Fidalgo**, Instituto Superior Técnico, University of Lisbon. **2019 - 2020**
Co-Advisor with Joaquim Jorge.
Thesis: *MAGIC: Manipulating Avatars and Gestures to Improve Remote Collaboration*
[J7]

2 **Francisco Venda**, Instituto Superior Técnico, University of Lisbon. **2016 - 2017**
Unofficial Co-Advisor with Joaquim Jorge.
Thesis: *Safe Walking in VR*
[C14]

1 **João Vieira**, Instituto Superior Técnico, University of Lisbon. **2014 - 2015**
Unofficial Co-Advisor with Joaquim Jorge and Artur Arsénio.
Thesis: *SleeveAR: Augmented Reality for Rehabilitation Using Realtime Feedback*
[C4] [06]

Undergrad Researchers

6 **Chao-Jung Lai**, University of Toronto. **2023 - 2024**
Co-Advisor with Tovi Grossman and Ludwig Sidenmark.
[C33]

5 **Helena Jovic**, University of Toronto. **2022 - 2023**
Co-Advisor with Tovi Grossman.

4 **Sixuan Wu**, University of Toronto. **2022 - 2023**
Co-Advisor with Tovi Grossman.
[C23]

3 **Angela Yang**, University of Toronto. **2022**
Co-Advisor with Tovi Grossman.
[C24]

2 **Tianquan (Andy) Di**, University of Toronto. **2021 - 2022**
Co-Advisor with Tovi Grossman.
[013]

1 **Kevin Huang**, University of Toronto. **2020 - 2021**
Co-Advisor with Tovi Grossman.
[C20]

Research Assistants

1 **Chu Li**, University of Toronto. **2021 - 2022**
Co-Advisor with Tovi Grossman.
[C19]

Mentoring (Ph.D. Students)

4 **Jiannan Li**, University of Toronto. **2020 - 2023**
[C24] [C23] [C20] [C19] [C18] [010]
3 **Laura (Di) Chen**, University of Toronto. **2020 - 2023**
2 **Fengyuan Zhu**, University of Toronto. **2020 - 2023**
[C29] [C27] [C21]
1 **Karthik Mahadevan**, University of Toronto. **2020 - 2023**
[C24] [C16]

Funding

5 **Snap Creative Challenge**
Project: *Audio-Augmented Reminiscence by Capturing and Replaying Memories*
Team Member
2023
4 **Champalimaud Center for the Unknown Grant & INESC-ID**
Project: *Laparoscopic Surgery Through Augmented Reality*
A viability study for the inclusion of Augmented Reality in the laparoscopic surgical theater.
Researcher
2018
3 **Portuguese Science and Technology Foundation Grant**
Project: *Interactive Tablets for Collaborative Scenarios Related to 3D Medical Image Exploration*
Interactive experiences for medical collaborative workspaces 3D images.
Researcher
2017
2 **Portuguese Science and Technology Foundation Grant**
Project: *Digital Mockup: Touching the 3rd Dimension*
Interaction techniques for architectural 3D modeling design and review in mixed reality
Researcher
2016
1 **Portuguese Science and Technology Foundation Grant**
Project: *CEDAR - Collaborative Engineering Design And Review*
Design and evaluation of novel collaborative spatial interaction techniques for the Oil & Gas Industry
Researcher
2015

Invited Talks

3 *Towards Intelligent Spaces: From Tracking Bodies to Perceiving Intent*, Instituto Superior Tecnico, University of Lisbon, April 2026
Host: Professor Joaquim Jorge
2 *Sketching Embodied Interactions: How to Design and Communicate User Experiences*, Keio University Graduate School of Media Design, Tokyo. November, 2025
Guest lecture for "Embodied Interactions"
Host: Professor Kouta Minamizawa
1 *From Passive Trackers to Proactive AI Wearable Companions: Uniting Nonverbal Communication and Spatial Understanding*, Google, Seattle. June, 2025
Host: Mar Gonzalez-Franco, Ph.D.

Lectures

Course Lectures

9 Media Design Research 1. *Research Methodology - Quantitative Research*. KMD. Fall 2026
8 Innovation Pipeline 1, *Prototyping Interactive Experiences #2: Vibecoding is the new Prototyping*. Fall 2026
7 Innovation Pipeline 1, *Sketching User Experiences 2: How to Design and Communicate User Experiences (Advanced)*. Fall 2026
6 Innovation Pipeline 1, *Sketching User Experiences 1: How to Design and Communicate User Experiences (Fundamentals)*. Fall 2026
5 Innovation Pipeline 2, *Prototyping Interactive Experiences #2: Vibecoding is the new Prototyping*. Spring 2026.
4 Innovation Pipeline 2, *Sketching Interactive Experiences*. Spring 2026.
3 Innovation Pipeline 2, *Team Building & Ideation*, (with Kouta Minamizawa and Ximing Elizabeth Shen). Spring 2026.
2 Media Design Research 1. *Research Methodology - Quantitative Research*. KMD. Spring 2026
1 Innovation Pipeline 2, *DESIGN & ENGINEERING: From Sketch to Prototype*, KMD, Fall 2025.

Seminar and Special Lectures

6 *KMD Crash Course. (IMPACT) Final Workshop: Future Ideas Showcase.* KMD. Fall 2026.
5 *KMD Crash Course. (MAKE) Prototyping Workshop: Sketching and Crafting.* KMD. Fall 2026.
4 *KMD Crash Course. (REFRAME) Introduction to the KMD Crash Course.* KMD. Fall 2026.
3 *KMD Crash Course. (MAKE) Prototyping Workshop: Sketching and Crafting.* KMD. Spring 2026.
2 *How to plan, write, and submit an ACM UIST research paper.* KMD. 2026.
1 *Introduction to ACM SIGGRAPH and ACM SIGCHI.* KMD. 2025.

Teaching Assistant

Human-Computer Interaction **2017/18**
Instituto Superior Técnico, University of Lisbon
Undergraduate course on the fundamental principles and rules for the design, development, and evaluation of interactive devices, systems, and services.

Awards and Recognition

14 Two Special Recognitions for Outstanding Reviews
CHI 2026 Papers
13 One Special Recognition for Outstanding Reviews
CHI 2025 Papers
12 Recognition for Efforts as a Peer Reviewer
IEEE TVCG 2024
11 Best Paper Award ACM VRST 2024
[C30] *Exploring User Placement for VR Remote Collaboration in a Constrained Passenger Space.*
10 Jazzki 2024 Human-Machine Creativity Award, First Place
[C28] *SynthScribe: Deep Multimodal Tools for Synthesizer Sound Retrieval and Exploration.*
9 One Special Recognition for Outstanding Reviews
UIST 2023 Papers
8 Recognition of Service Award, IEEE VR 2023.
In appreciation for contributions to the IEEE VR Conference as Videos Co-Chair.
7 Special Recognition for Outstanding Reviews
UIST 2022 Papers
6 Best Paper Honorable Mention Award ACM CHI 2022
[C20] *immersivePOV: Filming How-To Videos with a Head-Mounted 360° Action Camera.*
5 Recognition of Service Award IEEE VR 2022.
In appreciation for contributions to IEEE VR Conference as Web Chair.
4 Best Paper Honorable Mention Award ACM CHI 2021.
[C16] *"Grip-that-there": An Investigation of Explicit and Implicit Task Allocation Techniques for Human-Robot Collaboration.*
3 Paper featured in the 'Best of CHI 2019' event by IndiaHCI.
[C12] *WARPING DEIXIS: Distorting Gestures to Enhance Collaboration.*
2 Paper recipient of the Encarnação Award 2020 from Eurographics Portuguese chapter.
[J4] *Magic Carpet: Interaction Fidelity for Flying in VR.*
1 Honorable Mention Award in the 33rd Spring Conference on Computer Graphics 2017.
[J3] *Design and evaluation of novel out-of-reach selection techniques for VR using iterative refinement.*

Selected Media

2 Tuning into Tomorrow. AI can help musicians compose and create new sounds.

Is it just another music-making tool – or something else?
Adina Bresge, University of Toronto Magazine. April 2024
[C28]

1 Interactive 'Stargazer' camera robot assists with how-to video creation.
Krystle Hewitt, Department of Computer Science, University of Toronto. May 2023
[C24]

Professional Membership

ACM (Association for Computing Machinery) - https://www.acm.org
Professional Member **since 2021**
Student Member **2016 - 2021**

Contact References

Contact references available upon request.